

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical – II

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PART – A

Write the program with different features of C#

- a) Function Overloading
- b) Inheritance (all types)
- c) Constructor overloading
- d) Interfaces
- e) Using Delegates and events
- f) Exception handling

PART B :

Solution:

A) CODE

```
using System;

public class ABC
{
    void Sun1(int num1, int num2)
    {
        Console.WriteLine("Subtraction of Two Numbers : " + (num1 - num2));
    }

    void Sun1(int num, int num1, int num2)
    {
        Console.WriteLine("Addition of 3 Numbers : " + (num + num1 + num2));
    }

    public static void Main(string[] args)
    {
        ABC obj = new ABC();
    }
}
```

```

        obj.Sun1(68, 1);
        obj.Sun1(11, 12, 13);
    }
}

```

OUTPUT :-

```

Subtraction of Two Numbers : 67
Addition of 3 Numbers : 36

```

B) CODE :-

```

using System;

public class ABC
{
    ABC(int num1, int num2)
    {
        Console.WriteLine("Subtraction of Two Numbers : " + (num1 - num2));
    }

    ABC(int num, int num1, int num2)
    {
        Console.WriteLine("Addition of 3 Numbers : " + (num + num1 + num2));
    }

    public static void Main(string[] args)
    {
        ABC obj = new ABC(66, 7);
        ABC obj1 = new ABC(11, 12, 13);
    }
}

```

OUTPUT :-

```

Subtraction of Two Numbers : 59
Addition of 3 Numbers : 36

```

C) - SINGLE INHERITANCE CODE :-

```

using System;

public class A

```

```

{
    public void Fun1()
    {
        Console.WriteLine("Method of Class A");
    }
}

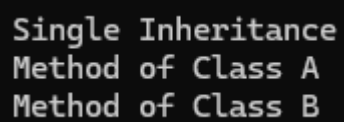
public class B : A
{
    public void Fun2()
    {
        Console.WriteLine("Method of Class B");
    }
}

class SingleInheritance
{
    public static void Main(string[] args)
    {
        Console.WriteLine("Single Inheritance");

        B obj = new B();
        obj.Fun1();
        obj.Fun2();
    }
}

```

OUTPUT :-



```

Single Inheritance
Method of Class A
Method of Class B

```

- MULTILEVEL INHERITANCE CODE :-

```

using System;

public class A
{

```

```

    public void Fun1()
    {
        Console.WriteLine("Method of Class A");
    }
}

public class B : A
{
    public void Fun2()
    {
        Console.WriteLine("Method of Class B");
    }
}

public class C : B
{
    public void Fun3()
    {
        Console.WriteLine("Method of Class C");
    }
}

class MultilevelInheritance
{
    public static void Main(string[] args)
    {
        Console.WriteLine("Multilevel Inheritance");
        C obj = new C();
        obj.Fun1();
        obj.Fun2();
        obj.Fun3();
    }
}

```

OUTPUT :-

Multilevel Inheritance
Method of Class A
Method of Class B
Method of Class C

- **HIERARICAL INHERITANCE CODE :-**

```
using System;
public class A
{
    public void Fun1()
    {
        Console.WriteLine("Method of Class A");
    }
}
public class B : A
{
    public void Fun2()
    {
        Console.WriteLine("Method of Class B");
    }
}
public class C : A
{
    public void Fun3()
    {
        Console.WriteLine("Method of Class C");
    }
}
class HierarchicalInheritance
{
    public static void Main(string[] args)
    {
        Console.WriteLine("Hierarchical Inheritance");
        B obj1 = new B();
        obj1.Fun1();
        obj1.Fun2();

        C obj2 = new C();
        obj2.Fun1();
        obj2.Fun3();
    }
}
```

OUTPUT :-

```
Hierarchical Inheritance
Method of Class A
Method of Class B
Method of Class A
Method of Class C
```

D) CODE :-

```
using System;

public interface A
{
    void Fun1();
}

public interface B
{
    void Fun2();
}

public class C : A, B
{
    public void Fun1()
    {
        Console.WriteLine("Method of Interface A");
    }

    public void Fun2()
    {
        Console.WriteLine("Method of Interface B");
    }
}

class InterfaceDemo
{
    public static void Main(string[] args)
    {
        C obj = new C();

        obj.Fun1();
        obj.Fun2();
    }
}
```

OUTPUT :-

```
Method of Interface A
Method of Interface B
```

E) CODE :-

```
using System;
```

```

class Program
{
    // Delegate
    public delegate void Message();

    // Event
    public static event Message ShowMessage;

    static void Main()
    {
        ShowMessage += Display;
        ShowMessage();
    }

    static void Display()
    {
        Console.WriteLine("Hello! Delegate and Event Example.");
    }
}

```

OUTPUT :-

```

Hello! Delegate and Event Example.

```

F) CODE :-

```

using System;
class Program
{
    static void Main()
    {
        try
        {
            int a = 10;
            int b = 0;
            int c = a / b;

            Console.WriteLine(c);
        }
        catch (DivideByZeroException)
        {
            Console.WriteLine("Cannot divide by zero.");
        }
    }
}

```

OUTPUT :-

```

Cannot divide by zero.

```